ORTHOGONAL ALIGNMENT OF TEACHER GEOMETRY FOR EMBEDDING DISTILLATION

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ABSTRACT

1 INTRODUCTION

2 RELATED WORK

3 METHOD

Let $f_T : \mathcal{X} \to \mathbb{R}^{d_T}$ be a frozen teacher and f_S a student with L Transformer layers and width $d_S < d_T$. Writing $\mathrm{norm}(\mathbf{x}) = \mathbf{x}/(\|\mathbf{x}\|_2 + \epsilon)$, we run the teacher once over an unlabeled corpus $\{x_i\}_{i=1}^N$ and cache only its final sentence embeddings, $\widetilde{\boldsymbol{\tau}}_i = \mathrm{norm}(f_T(x_i))$. Pooling and normalising the student's final token states gives $\mathbf{z}_i \in \mathbb{S}^{d_S-1}$. The two vectors have different widths, so nothing can be compared until a map bridges them; we fix that map before training and never touch it again.

3.1 TEACHER TARGETS

We fit the leading d_S-dimensional principal subspace $P_{\mathrm{PCA}} \in \mathbb{R}^{d_T \times d_S}$ on the centred cache. Its orientation inside that subspace is arbitrary — invisible to every cosine-metric evaluation, but not to a point-wise loss — so we remove it once by orthogonal Procrustes against the *untrained* student. With $\mathbf{T}_{\mathrm{PCA}}$ the normalised principal coordinates of a fitting subset and $\mathbf{Z}_0$ the student's normalised outputs on the same sentences (Schönemann, 1966),

$$R^\star = \operatorname*{arg\,min}_{R \in O(d_S)} \|\mathbf{T}_{\mathrm{PCA}} R - \mathbf{Z}_0\|_F = UV^\top, \qquad U\Sigma V^\top = \mathrm{svd}\big(\mathbf{T}_{\mathrm{PCA}}^\top \mathbf{Z}_0\big)\,, \tag{1}$$

and the target of sentence x_i is

$$\boldsymbol{\tau}_i = \mathrm{norm}(f_T(x_i)\, P_{\mathrm{PCA}} R^\star) \in \mathbb{S}^{d_S-1}. \tag{2}$$

Both factors are fitted before the first gradient step and frozen for the whole run. Because $R^\star$ is orthogonal it leaves the targets' pairwise cosines and retained variance exactly unchanged, so the two factors are independent: the projection decides *what* of the teacher survives, the rotation only *where* it is placed relative to the student's starting point.

3.2 OBJECTIVE

Only the final student embedding is supervised. The endpoint term aligns each student embedding with its fixed, dimension-matched teacher target:

$$\mathcal{L}_{\mathrm{end}} = \frac{1}{B} \sum_{i=1}^{B} \left(1 - \langle \mathbf{z}_i, \boldsymbol{\tau}_i \rangle\right), \tag{3}$$

where B is the minibatch size. This term fixes point-wise correspondence in the gauge selected by $P_{\mathrm{PCA}} R^\star$.

The second term matches the topology of the two embedding clouds without requiring a shared ambient dimension. Let $\mathrm{Dgm}_k(\mathbf{X})$ denote the degree-k Vietoris–Rips persistence diagram of the row-wise normalised point cloud $\mathbf{X}$ under the chord distance $d(\mathbf{u},\mathbf{v}) = \sqrt{2 - 2\langle \mathbf{u},\mathbf{v}\rangle}$. For $k = 0$, the finite death times are exactly the $B - 1$ edge weights of the cloud's minimum spanning tree. Writing their sorted values as $\delta^S_{(j)}$ and $\delta^T_{(j)}$, we use

$$\mathcal{L}_{H_0} = \frac{1}{B-1} \sum_{j=1}^{B-1} \left(\delta^S_{(j)} - \delta^T_{(j)} \right)^2, \tag{4}$$

$$\mathcal{L}_{\mathrm{topo}} = \mathcal{L}_{H_0} + \lambda_1 \, W_2^2 \Big(\mathrm{Dgm}_1(\mathbf{Z}), \mathrm{Dgm}_1(\widetilde{\mathbf{T}}) \Big), \tag{5}$$

where W_2 permits low-persistence cycles to match to the diagonal. The teacher cloud $\widetilde{\mathbf{T}}$ contains the original cached embeddings $\widetilde{\boldsymbol{\tau}}_i \in \mathbb{R}^{d_T}$, not their projected targets. Consequently, $\mathcal{L}_{\mathrm{topo}}$ preserves connectivity and cycles from the teacher's own space and is independent of the endpoint interface. Setting $\lambda_1 = 0$ gives the H_0-only variant.

The complete training objective contains only these two losses:

$$\mathcal{L} = \lambda_{\mathrm{end}} \mathcal{L}_{\mathrm{end}} + \lambda_{\mathrm{topo}} \mathcal{L}_{\mathrm{topo}}. \tag{6}$$

The endpoint term transfers sample-wise semantics, while the topological term constrains batch-level structure that a point-wise objective does not explicitly measure.

What the method does not contain. No projector is trained, no contrastive or Gram loss is added, no intermediate layer is supervised, and nothing is attached to the student at inference: the deployed model is the unmodified encoder. Training costs one teacher pass over the corpus, one truncated SVD of the cache, one $d_S \times d_S$ SVD for $R^\star$, and then ordinary student training against the cached teacher embeddings and their fixed projected targets — no teacher hidden state, attention map or online teacher forward pass is used.

4 EXPERIMENTS

4.1 SETUP

Teacher–student pairs. We evaluate three teacher–student configurations that differ in teacher family, teacher width, and student capacity: (a) Qwen3-Embedding-4B ($d_T = 2560$, last-token pooling) $\rightarrow$ BERT-base (109M, $d_S = 768$, 12 layers), the capacity-gap setting; (b) BGE-M3 ($d_T = 1024$, an XLM-R encoder with CLS pooling) $\rightarrow$ MiniLMv2-L6-H768 (66M, $d_S = 768$, 6 layers), a different teacher family; (c) Qwen3-Embedding-0.6B ($d_T = 1024$, last-token pooling) $\rightarrow$ MiniLMv2-L6-H384 (22M, $d_S = 384$, 6 layers), the compact-student setting.

Training corpus. All methods are distilled on the same corpus of **102,361 unlabeled sentences**, sampled once from the training splits of the nine evaluation benchmarks (SciTail; Tweet; Emotion; STS-B; Banking77; MRPC; WiC; SICK; STS12) and fixed across every run in Table 1. Only raw text is read: no label, pair annotation, or similarity score enters any objective, so the training signal is unsupervised apart from the teacher.

Evaluation. We score frozen student embeddings on nine benchmarks in three families. For classification (Banking77, Emotion, Tweet) a logistic-regression probe is fitted on that benchmark's own training split and scored with macro-F1. For pair classification (MRPC, SciTail, WiC) the cosine similarity is mapped to $[0, 1]$ and scored with average precision. For STS (SICK, STS12, STS-B) we report Spearman correlation against gold scores. The Avg. column is the unweighted mean of the nine primary scores. Appendix **??** gives the split sizes and the threshold protocol.

Table 1: Experimental results on nine datasets. Avg. is the unweighted mean of the nine benchmark scores. Where shown, uncertainty is the sample standard deviation over three seeds.

	Classification (F1)			Pair Classification (AP)			STS (Spearman)			
Method	Banking77	Tweet	Emotion	MRPC	SciTail	WiC	SICK	STS12	STS-B	Avg.
(a) Qwen3-Embedding-4B → BERT-base (109M)										
Teacher	93.63	72.22	68.94	84.79	91.77	66.80	83.43	82.55	86.87	81.22
Student base	84.86	47.79	68.02	77.36	67.74	59.22	42.43	21.54	20.30	54.36
RKD	92.09 ± 0.13	72.31 ± 0.21	63.63 ± 0.19	85.02 ± 0.08	80.54 ± 0.59	66.43 ± 0.11	79.39 ± 0.08	71.96 ± 0.15	77.72 ± 0.08	76.56 ± 0.09
Stella	90.15 ± 0.10	72.05 ± 0.01	60.05 ± 0.64	85.96 ± 0.05	79.00 ± 0.36	**69.83** ± 0.15	76.66 ± 0.20	69.94 ± 0.28	77.22 ± 0.07	75.65 ± 0.07
CDM	-	-	-	-	-	-	-	-	-	-
DSKD	-	-	-	-	-	-	-	-	-	-
EMO	-	-	-	-	-	-	-	-	-	-
TALAS	92.19 ± 0.02	**75.11** ± 0.02	72.10 ± 0.17	**86.31** ± 0.02	87.00 ± 0.14	67.99 ± 0.06	80.24 ± 0.04	**76.98** ± 0.10	82.43 ± 0.06	80.04 ± 0.05
GeoODE-KD	**93.53** ± 0.10	74.61 ± 0.54	**73.12** ± 0.52	85.43 ± 0.07	**87.14** ± 0.14	66.39 ± 0.08	**81.89** ± 0.03	76.05 ± 0.17	**82.49** ± 0.09	**80.07** ± 0.08
(b) BGE-M3 → MiniLMv2-H768 (66M)										
Teacher	93.52	73.85	68.56	85.81	91.87	61.47	79.18	78.73	84.87	79.76
Student base	81.84	47.70	71.67	79.67	65.06	60.89	49.46	26.76	24.12	56.35
RKD	91.60 ± 0.19	74.61 ± 0.16	63.04 ± 0.73	85.83 ± 0.10	79.72 ± 0.15	63.69 ± 0.14	76.77 ± 0.11	69.14 ± 0.18	75.46 ± 0.09	75.54 ± 0.07
Stella	-	-	-	-	-	-	-	-	-	-
CDM	87.64 ± 0.06	70.90 ± 0.27	56.72 ± 0.56	85.65 ± 0.05	67.64 ± 0.31	**66.85** ± 0.14	70.26 ± 0.47	66.73 ± 0.28	69.40 ± 0.18	71.31 ± 0.13
DSKD	-	-	-	-	-	-	-	-	-	-
EMO	-	-	-	-	-	-	-	-	-	-
TALAS	92.12 ± 0.02	75.48 ± 0.23	65.32 ± 0.72	**87.05** ± 0.06	85.39 ± 0.13	64.40 ± 0.08	78.20 ± 0.05	71.49 ± 0.14	**81.05** ± 0.09	77.83 ± 0.05
GeoODE-KD	**92.58** ± 0.04	**76.40** ± 0.06	**66.90** ± 0.25	86.55 ± 0.05	**87.34** ± 0.05	63.31 ± 0.02	**79.23** ± 0.03	**72.72** ± 0.03	80.60 ± 0.13	**78.40** ± 0.04
(c) Qwen3-Embedding-0.6B → MiniLMv2-H384 (22M)										
Teacher	93.21	71.33	66.53	83.37	90.00	66.85	80.64	77.11	84.60	79.29
Student base	68.29	41.07	69.90	78.39	64.56	59.58	47.60	21.95	22.08	52.60
RKD	67.41	69.87	44.50	78.38	64.56	59.58	47.60	21.95	22.07	52.88
Stella	84.36	68.81	54.01	81.61	72.69	68.53	65.30	61.02	64.70	69.00
CDM	85.16 ± 0.08	69.12 ± 0.18	52.51 ± 0.21	83.98 ± 0.09	67.97 ± 0.36	66.55 ± 0.09	69.19 ± 0.19	62.92 ± 0.14	66.50 ± 0.07	69.32 ± 0.02
DSKD	85.00 ± 0.04	69.19 ± 0.32	52.77 ± 0.31	84.44 ± 0.06	67.44 ± 0.17	65.95 ± 0.03	69.01 ± 0.21	63.59 ± 0.07	66.41 ± 0.08	69.31 ± 0.04
EMO	84.40	69.20	56.13	81.68	72.21	68.22	63.82	57.88	63.34	68.54
TALAS	89.85 ± 0.10	73.74 ± 0.18	63.02 ± 0.14	**85.96** ± 0.01	82.02 ± 0.08	65.79 ± 0.06	75.70 ± 0.04	**73.83** ± 0.08	78.28 ± 0.07	76.47 ± 0.01
GeoODE-KD	**91.69** ± 0.13	**74.36** ± 0.18	**66.76** ± 0.37	84.99 ± 0.06	**83.53** ± 0.01	67.35 ± 0.04	**77.98** ± 0.06	72.20 ± 0.01	**78.32** ± 0.04	**77.47** ± 0.04

5 Analysis

6 Conclusion

References

Peter H. Schönemann. A generalized solution of the orthogonal Procrustes problem. *Psychometrika*, 31(1):1–10, 1966.